

From pixels to *the full picture*.

Satellite analytics, large geospatial datasets and machine learning for deforestation, crops and carbon, mapped with a cartographer's eye.

ABOUT

From paper maps and cadastre to satellite time series and foundation models.

I grew up in Porto and now live in Sofia. I've drawn cartographic and cadastral maps, assessed environmental impacts, and spent the last several years building crop analytics, learning how crops behave across every continent and season.

Today I help organisations turn satellite imagery, open and commercial, into evidence they can act on.

HOW I CAN HELP

Environmental due diligence

Deforestation screening against the EUDR cutoff, protected areas, soil and climate exposure.

Carbon & soil MRV

Practice evidence such as green cover between crops; persistent land-use change.

Crop & field monitoring

Crop type, phenology, anomalies and declaration checks, from one farm to a country.

Cartography

Maps designed to be understood, for reports, publications and print.

PATH

- **Cartography & cadastre**
where every boundary had to be right
- **Environmental assessment**
impact studies and baselines
- **Agtech**
several years of crop analytics at scale
- **Independent**
satellite analytics for environment & agriculture

WORKED WITH

Multinational food, beverage and retail companies, agricultural and forestry producers in Brazil, agricultural insurers, and the World Health Organization.

TOOLS

Python (xarray, dask, geopandas, scikit-learn, PyTorch) · QGIS · Google Earth Engine · STAC / COG

DATA

Sentinel-1/2/3/5P · Landsat · Maxar · Planet · LPIS · Copernicus DEM

LANGUAGES

English · Portuguese · Bulgarian · Spanish

BASED IN

Sofia, Bulgaria

Have land you need to understand?

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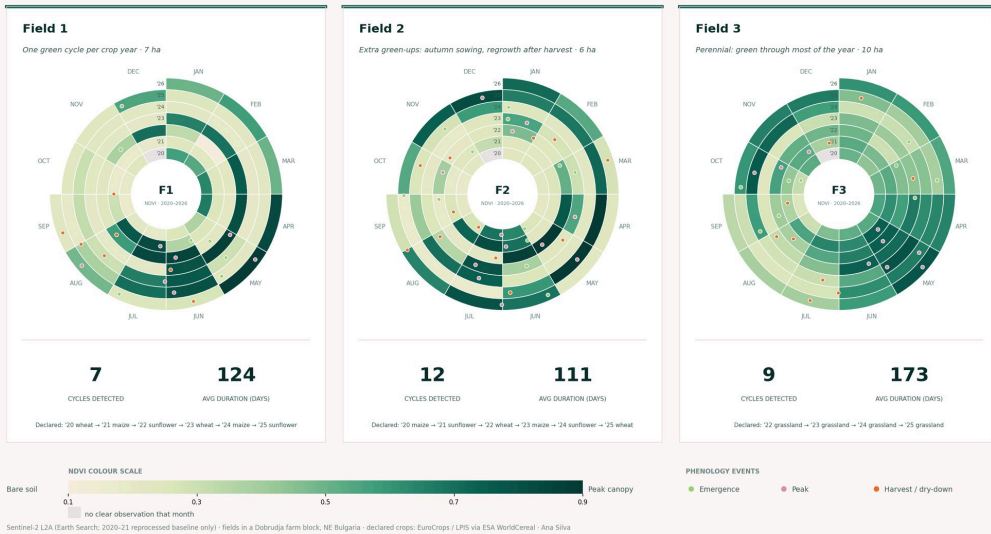
Selected work

Real Sentinel-1 and Sentinel-2 imagery · public parcel registers



Each ring is one year · each wedge one month · colour = that month's median NDVI · dots mark detected emergence, peak and harvest/dry-down.

JAN 2020 - SEP 2026
SENTINEL-2 L2A - NDVI
20% AMPLITUDE METHOD

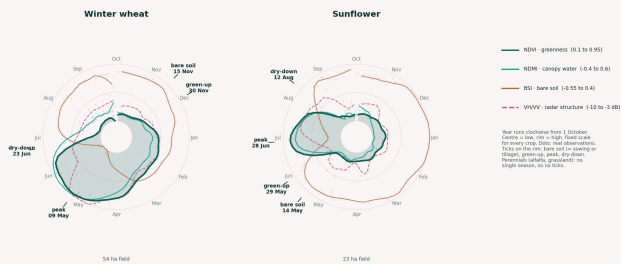


Phenology clock, seven years ring by ring

One ring per year on three neighbouring fields: rotations (wheat → maize → sunflower), emergence, peak and harvest, from 20,207 Sentinel-2 observations, 2020–2026.

Phenological clock: a field's year at a glance

Winter wheat greens in spring and is harvested by early summer; sunflower is sown in spring and peaks in summer. Real Sentinel-2 and Sentinel-3, Oct 2024 – Sep 2025.



Sentinel-2 L2A (Earth Search) · Sentinel-2 RTC (Planetary Computer) · Soil: LPIS 2023 via ESA WorldCereal · events approximate, from smoothed NDVI and BSI (dry down = greenness falls; harvest follows) · Ana Silva

A season as a clock

Sowing, green-up, peak and dry-down from Sentinel-2 and Sentinel-1. Every crop keeps its own time.

Crop greenness anomaly, 2020: peak-season NDVI vs the 2018-2025 normal
Baragan Plain, SE Romania (Danube cereals) - Sentinel-2, cropland only



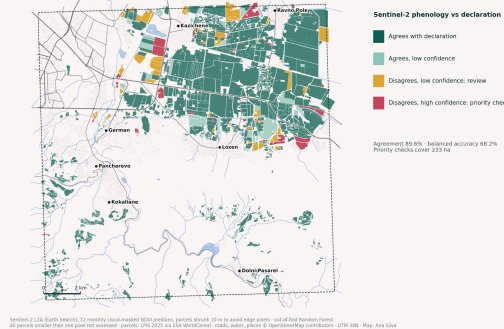
Brown = below-normal peak greenness, a yield risk proxy (not measured yield). 72% of scored cropland below -1 sd in 2020. Peak = max of monthly medians Apr-Aug; peaks need a 4 clear months. Sentinel-2 L2A (Earth Search, one processing version per scene) / cropland: Impact Observatory LLC 2021 / UTM 35N / Map: Jau Silva

Drought from space

2020 on the Danube plain: peak greenness 0.60 vs 0.76–0.85 in other years; 72% of cropland below normal.

Do declared crops match what the satellite sees?

CAP area-monitoring markers - 1,339 declared parcels SE of Sofia - 2024-25 crop year

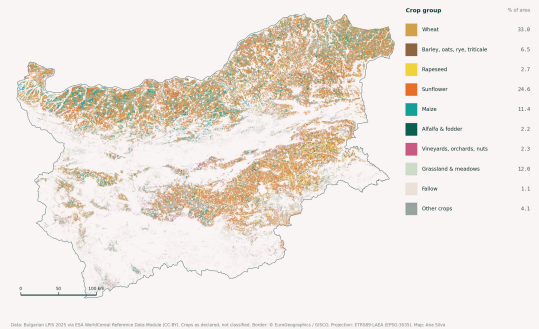


CAP area monitoring

1,339 parcels: 89.6% agreement, 68.2% balanced accuracy,
60 priority checks.

Every declared field in Bulgaria, 2025

1,122,696 parcels from the public land-parcel register (LPIS), coloured by the crop farmers declared



Data: Bulgarian UFS 2023 via ESA WorldCereal Reference Data Module (CC-BY). Crops as declared, not classified. Border: © EuroGeographics / GISCOP. Projection: ETRS89-LAEA (EPSG:31466). Map: Ana Silva

A country, field by field

1,122,696 declared parcels in Bulgaria, 2025, by crop.